Lab no. 2

Hands-On Git and Version Control

NAME: SANIA SHAFIQUE

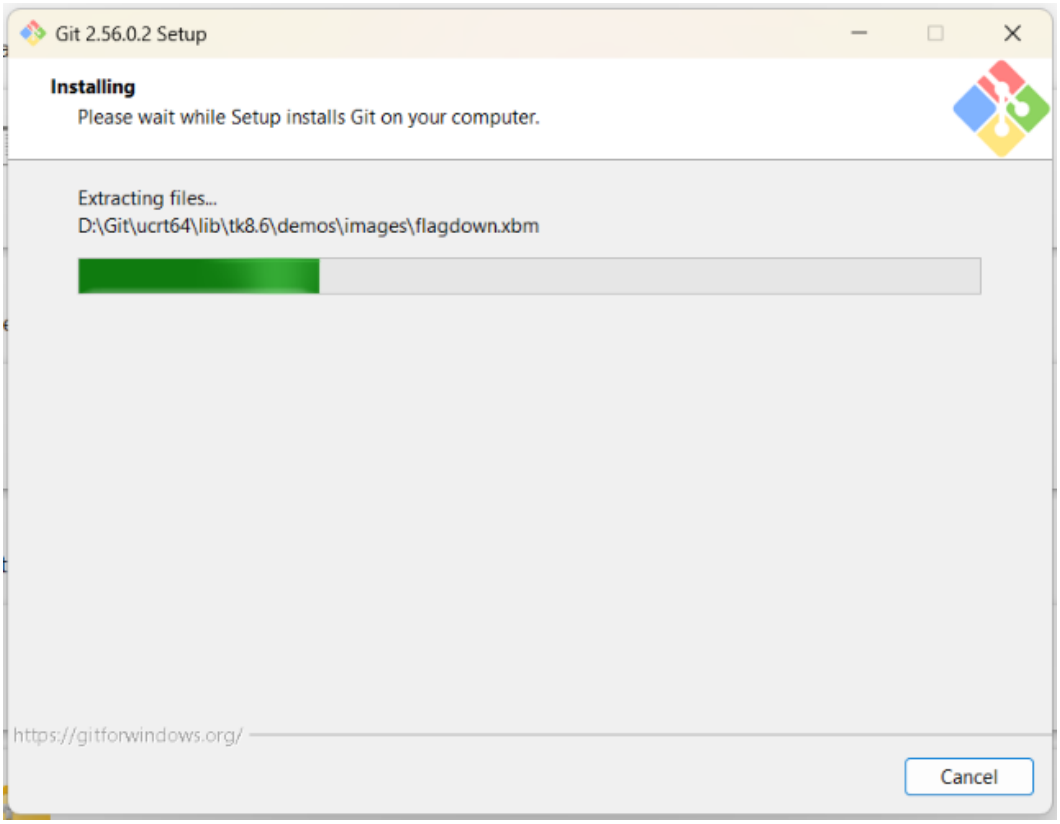
REGISTRATION NO.: 2023-BCS-078

COURSE CODE: CS-649

SECTION: A

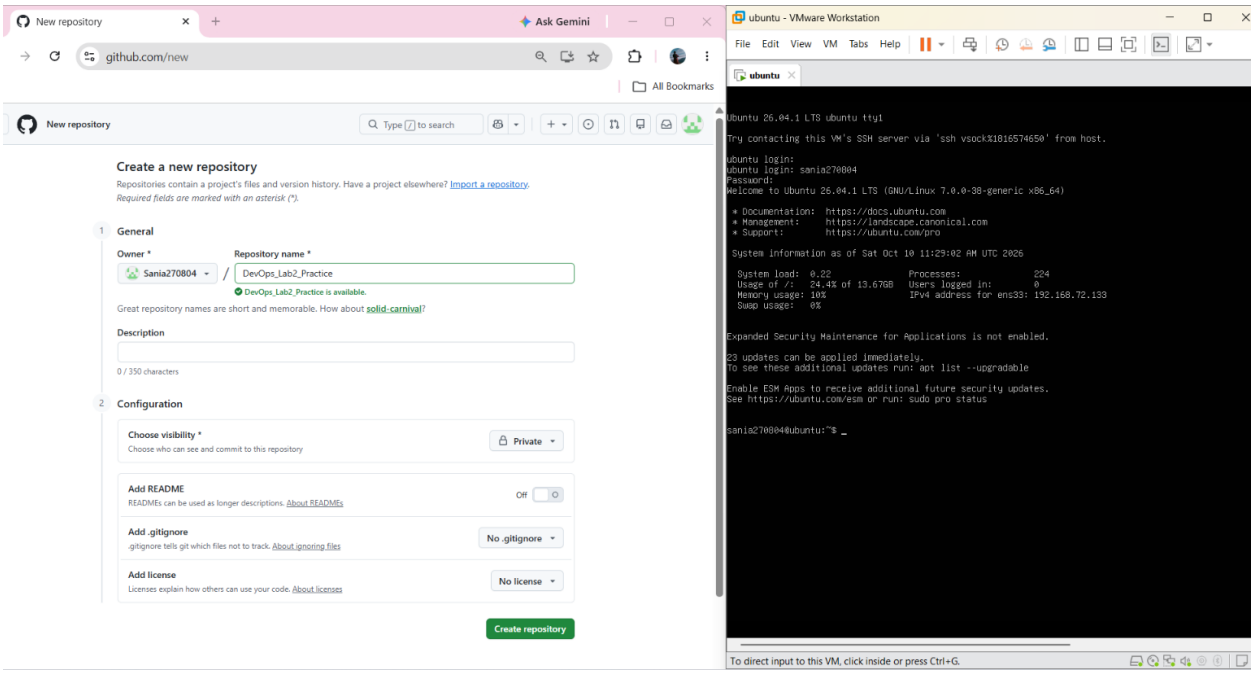
GITHUB USERNAME: Sania270804

Getting Started



Install git on your PC using the link given in the lab.

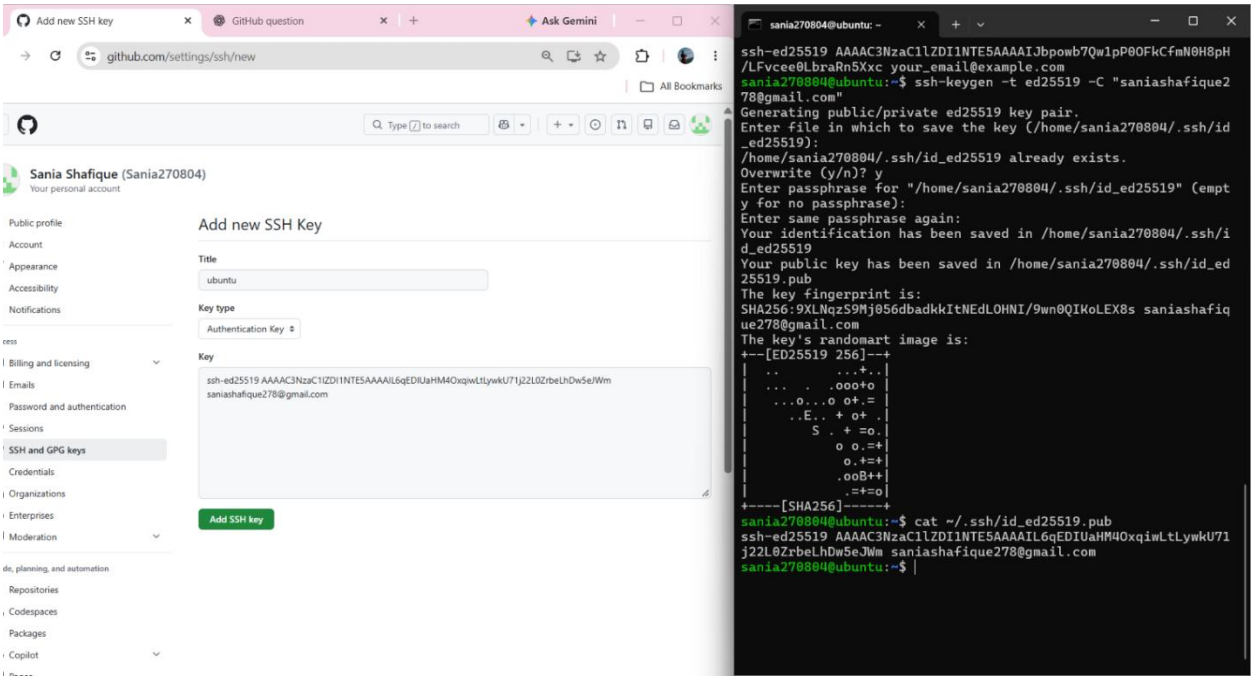
Task 1: Create Private GitHub Repository



Create a private repository named DevOps_Lab2_Practice.
Task 2: Connect Repository via SSH

```
sania270804@ubuntu:~$ ssh-keygen -t ed25519 -C "saniashafique278@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/sania270804/.ssh/id_ed25519):
/home/sania270804/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase for "/home/sania270804/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/sania270804/.ssh/id_ed25519
Your public key has been saved in /home/sania270804/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:9XLNqzS9Mj056dbadkkItNEdLOHNI/9wn0QIKoLEX8s saniashafique278@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|  ..       ...+..|
|  ...   .  .ooo+o |
|   ...o...o o+.= |
|     ..E.. + o+ .|
|        S . + =o.|
|           o o.=+|
|            o.+=+|
|           .ooB++|
|            .=+=o|
+----[SHA256]-----+
sania270804@ubuntu:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIL6qEDIUaHM4OxqiwLtLywkU71j22L0ZrbeLhDw5eJWm saniashafique278@gmail.com
```

Generate a new ssh key using the commands given in the lab.



Add that ssh key to your github account.

```
sania270804@ubuntu:~$ git clone git@github.com:sania270804/Dev
Ops_Lab2_Practice.git
Cloning into 'DevOps_Lab2_Practice'...
warning: You appear to have cloned an empty repository.
sania270804@ubuntu:~$
```

Clone your repository using ssh.
Task 3: Configure Git Username and Email

```
sania270804@ubuntu:~$ git config --global user.name "Sania Shafique"
git config --global user.email "saniashafique@gmail.com"
sania270804@ubuntu:~$
```

Set up your Git identity

```
sania270804@ubuntu:~$ git config --list
user.name=Sania Shafique
user.email=saniashafique@gmail.com
sania270804@ubuntu:~$
```

Verify your configuration
Task 4: Explore the .git Folder

```
sania270804@ubuntu:~$ cd DevOps_Lab2_Practice
sania270804@ubuntu:~/DevOps_Lab2_Practice$ ls -a .git
.  ..  HEAD  config  description  hooks  info  objects  refs
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Show hidden files and locate the .git directory.
Task 5: Local Repository Management

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ rm -rf .git
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git status ==> error why?
fatal: not a git repository (or any of the parent directories): .git
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Delete the existing .git folder from your cloned repo.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/sania270804/DevOps_Lab2_Practice/.git/
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Re-initialize the local git repository.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ echo "# Lab02 Git Practice" >> README.md
git add README.md
git commit -m "Initial commit"
[master (root-commit) 8e102a2] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Add a file named README.md and commit it.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git remote add origin git@github.com:sania270804/DevOps_Lab2_Practice.git
git branch -M main
git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 237 bytes | 118.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab2_Practice.git
To github.com:sania270804/DevOps_Lab2_Practice.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Connect your local repo to GitHub and push.
Task 6: File Status & Staging

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ echo "Adding some test notes" >> notes.txt
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        error
        notes.txt

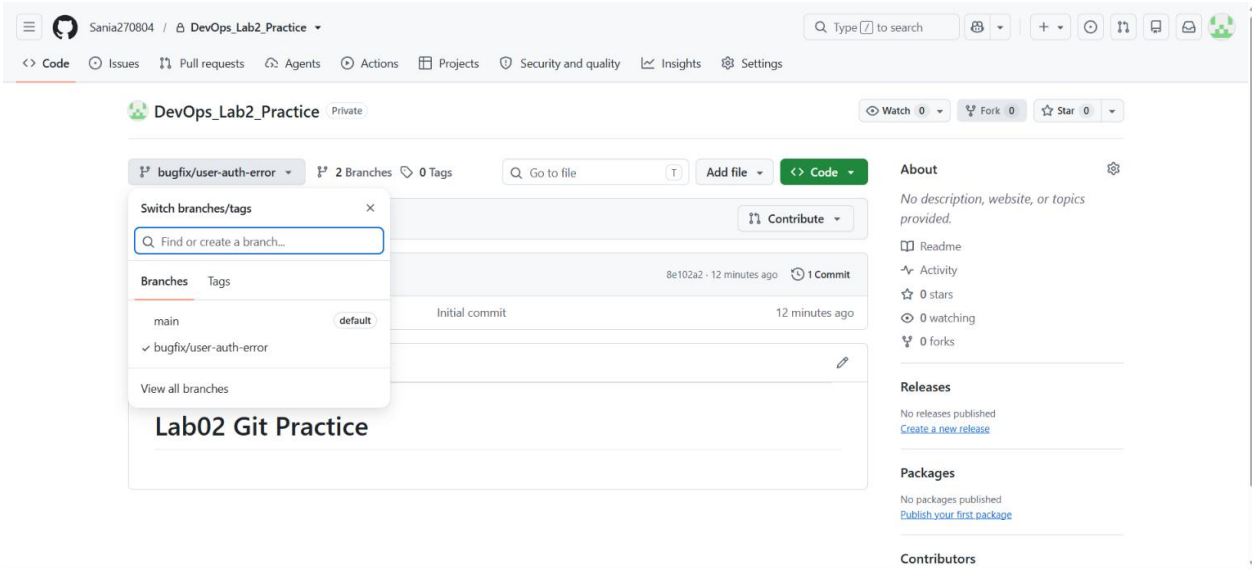
nothing added to commit but untracked files present (use "git add" to track)
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Create a new file notes.txt and write a note then check status.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
git commit -m "Add notes.txt"
[main e035e77] Add notes.txt
 1 file changed, 1 insertion(+)
 create mode 100644 notes.txt
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Stage and commit.

Task 7: Branch Creation Using GitHub GUI



On GitHub (web interface), create a branch named bugfix/user-auth-error.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git pull origin bugfix/user-auth-error
From github.com:sania270804/DevOps_Lab2_Practice
 * branch            bugfix/user-auth-error -> FETCH_HEAD
 * [new branch]      bugfix/user-auth-error -> origin/bugfix/user-auth-error
Already up to date.
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Pull the branch to your local repository to sync.

Task 8: Branch Creation and Push Using Git Bash

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature/db-connection
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 303 bytes | 303.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab2_Practice.git
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:      https://github.com/Sania270804/DevOps_Lab2_Practice/pull/new/feature/db-connection
remote:
To github.com:sania270804/DevOps_Lab2_Practice.git
 * [new branch]      feature/db-connection -> feature/db-connection
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Create a branch named feature/db-connection and push the branch to the remote repository.

Task 9: Branching & Merging

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature-1
Switched to a new branch 'feature-1'
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Create and switch to a branch feature-1.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ echo 'def greet(): print("Hello from feature-1")' >> main.py
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git add main.py
git commit -m "Add new function to main.py"
[feature-1 5938414] Add new function to main.py
 1 file changed, 1 insertion(+)
 create mode 100644 main.py
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Modify main.py (add a function) and commit.

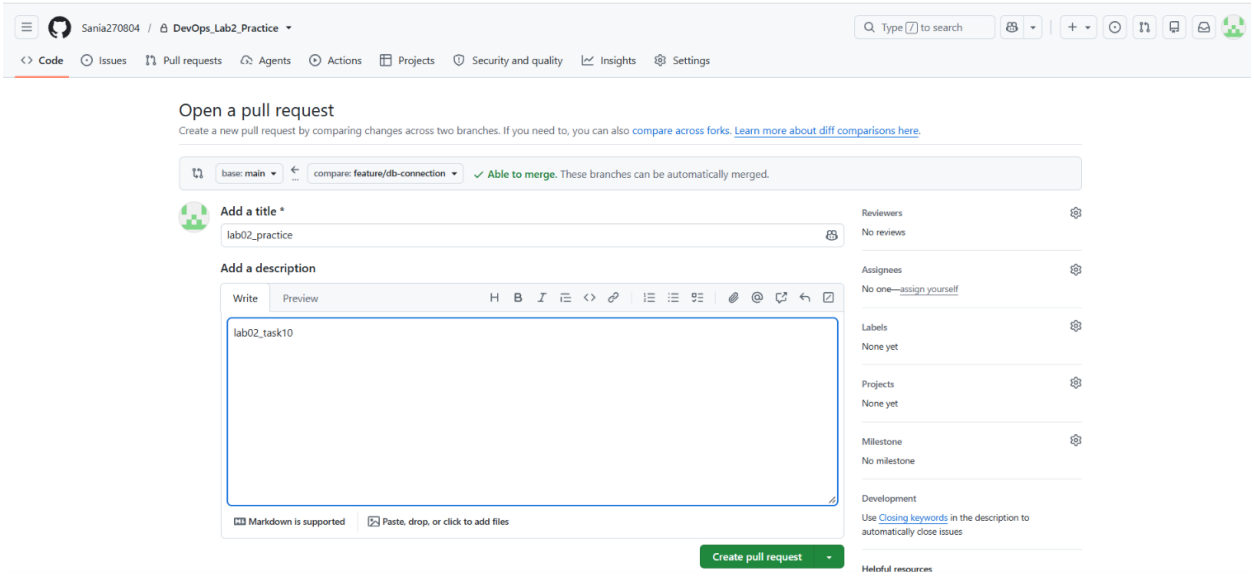
```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git merge feature-1
Updating e035e77..5938414
Fast-forward
 main.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 main.py
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Switch back to main and merge.

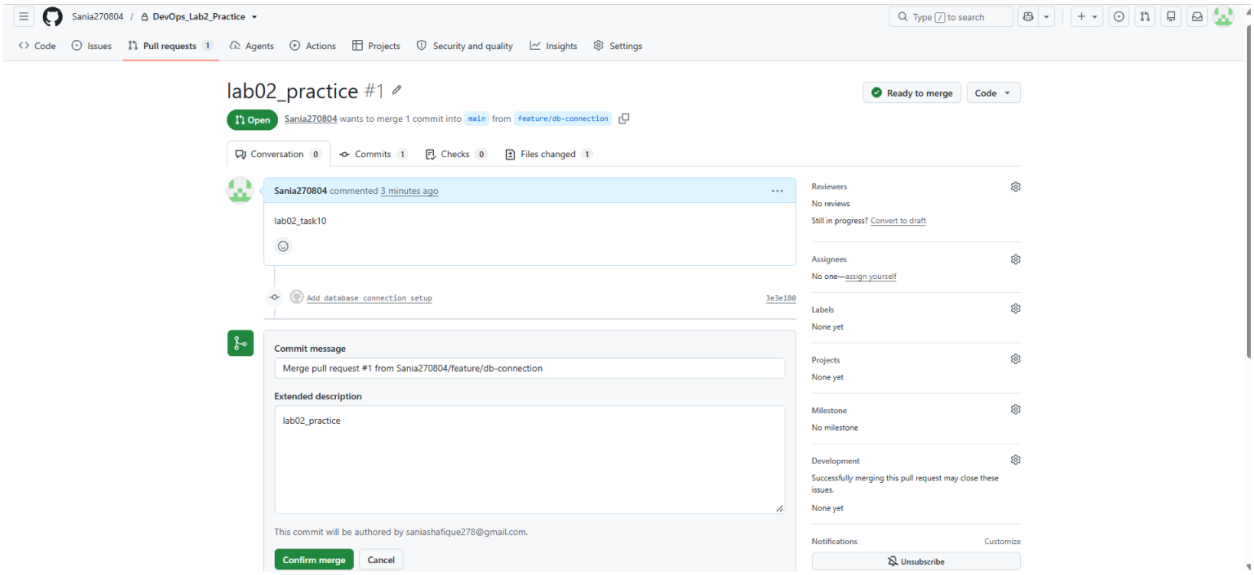
```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
git push origin feature-1
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 367 bytes | 183.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab2_Practice.git
To github.com:sania270804/DevOps_Lab2_Practice.git
   8e102a2..5938414  main -> main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab2_Practice.git
remote:
remote: Create a pull request for 'feature-1' on GitHub by visiting:
remote:      https://github.com/Sania270804/DevOps_Lab2_Practice/pull/new/feature-1
remote:
To github.com:sania270804/DevOps_Lab2_Practice.git
 * [new branch]      feature-1 -> feature-1
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Push all branches.

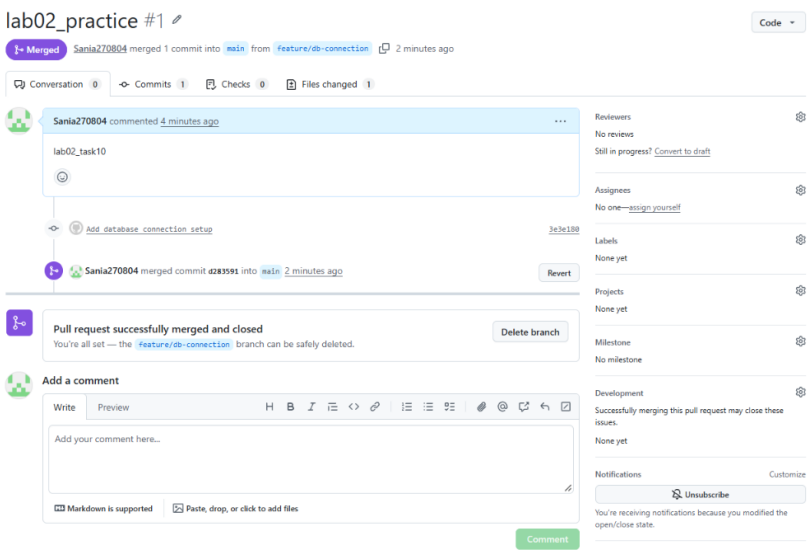
Task 10: Pull Request and Branch Review (GitHub GUI)



On GitHub, create a Pull Request from the branch feature/db-connection to main.



Review the Pull Request and merge it using the GitHub GUI.



After merging, delete the feature/db-connection branch using the GitHub GUI.
Task 11: Detailed Branch Strategy (Develop/Staging)

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git branch -a
  develop
  feature-1
  feature/db-connection
* feature/user-profile
  main
  staging
  remotes/origin/bugfix/user-auth-error
  remotes/origin/feature-1
  remotes/origin/feature/db-connection
  remotes/origin/main
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Create the branches (develop, staging, feature/*, bugfix/*) to simulate a professional branching strategy.

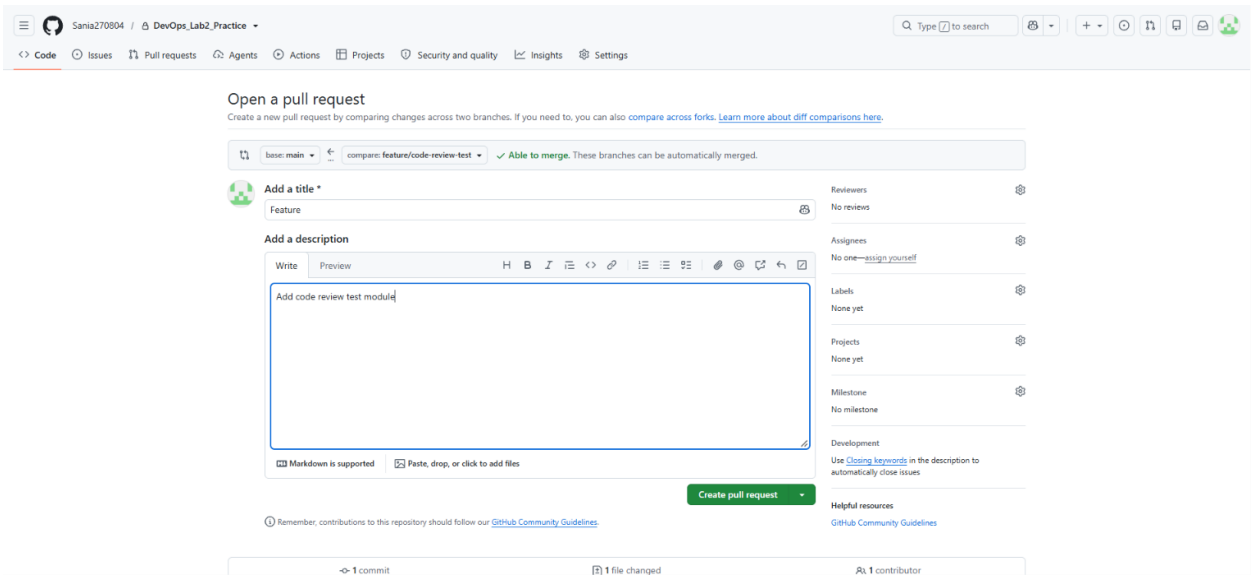
```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout staging
git merge develop
Switched to branch 'staging'
Updating d283591..847b0dd
Fast-forward
 profile.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 profile.py
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

develop is merged into staging for testing.

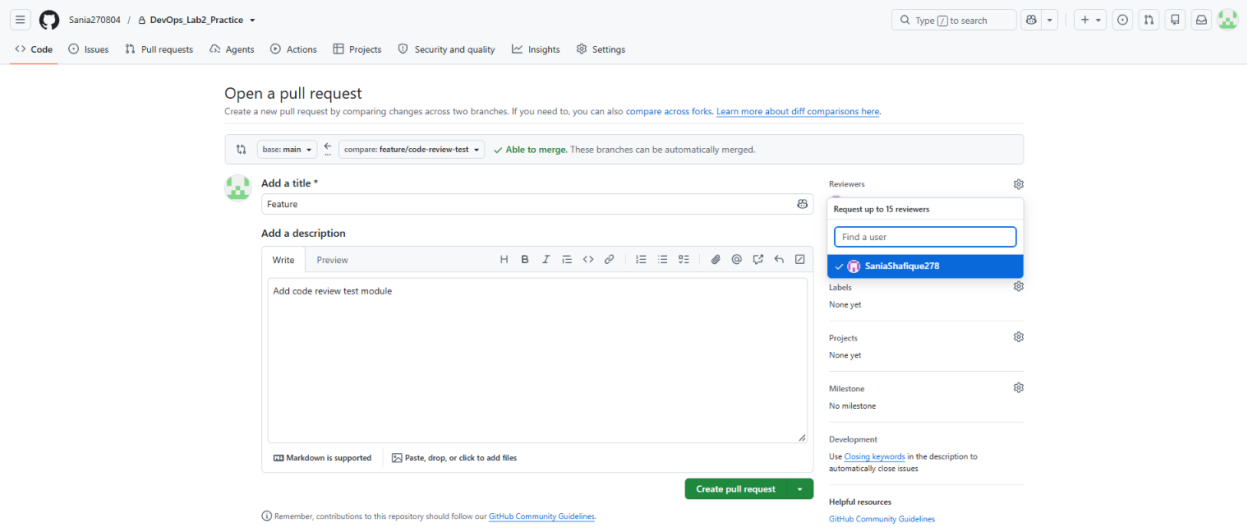
```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
git merge staging
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Updating d283591..847b0dd
Fast-forward
 profile.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 profile.py
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Finally, staging is merged into main (production).

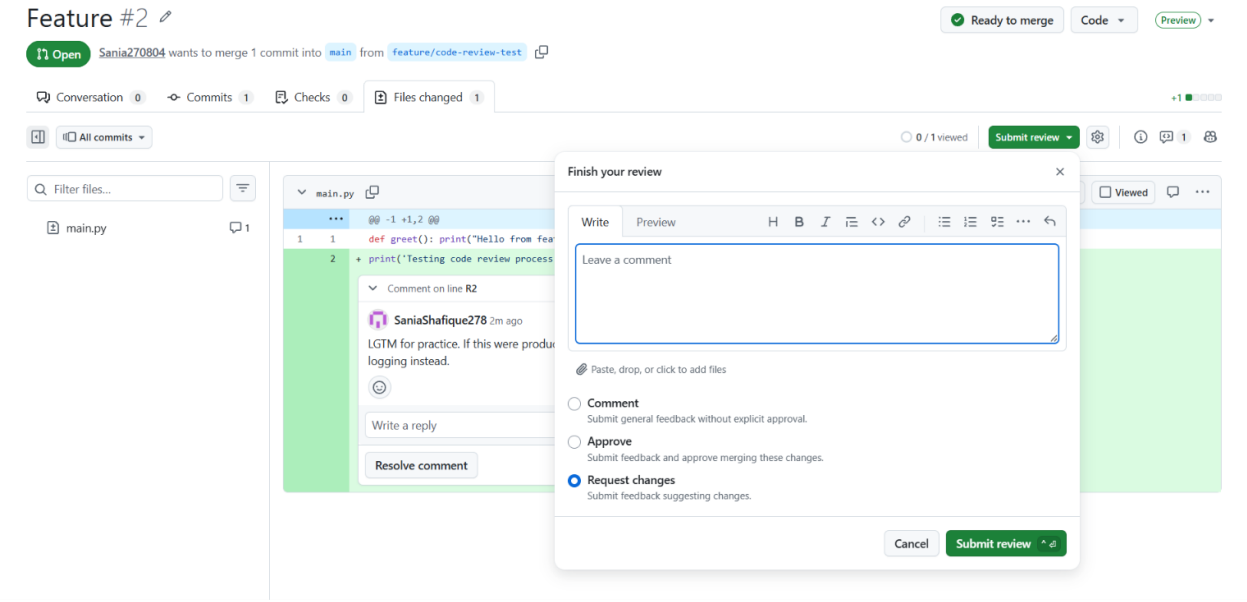
Task 12: Code Review Workflow



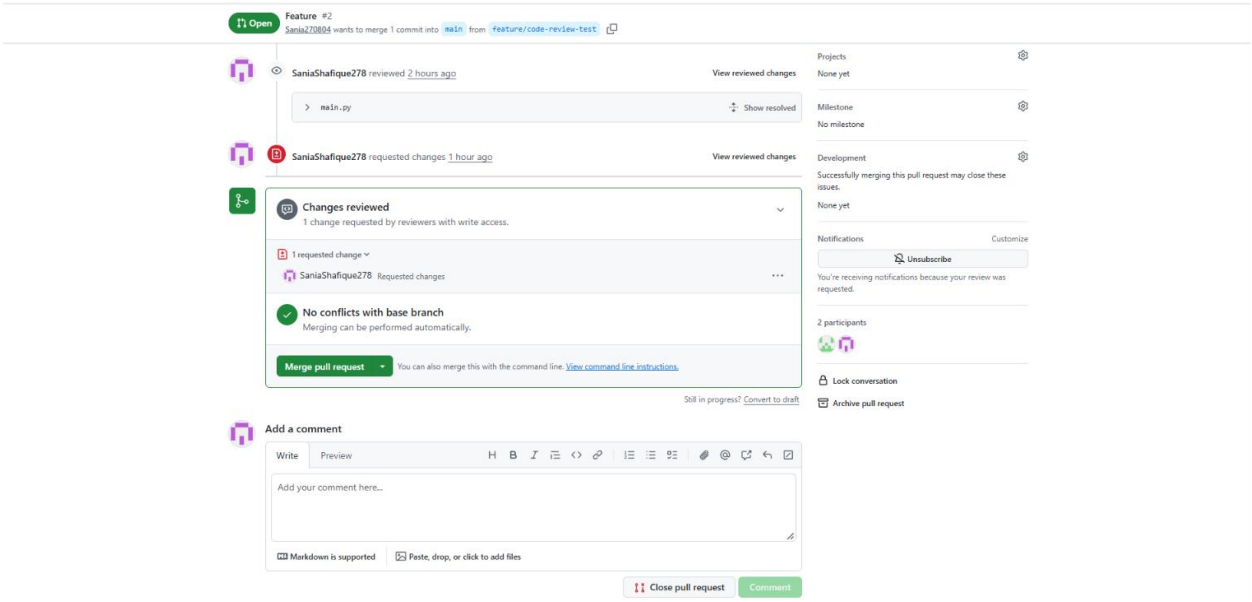
Create a Pull Request (PR).



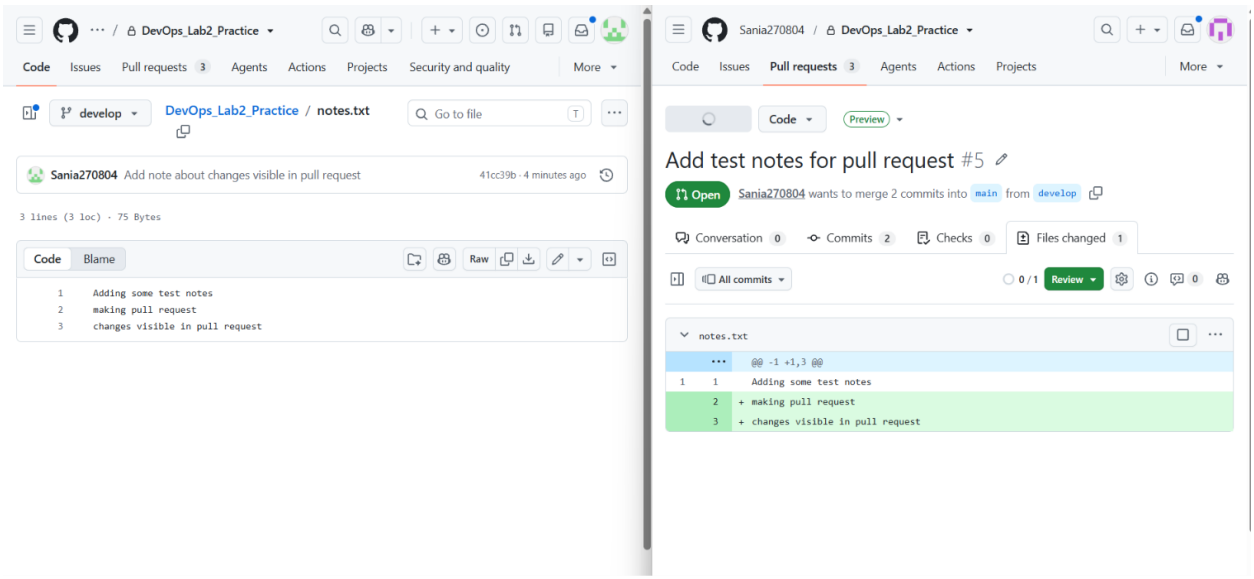
Assign a reviewer (teammate or second account).



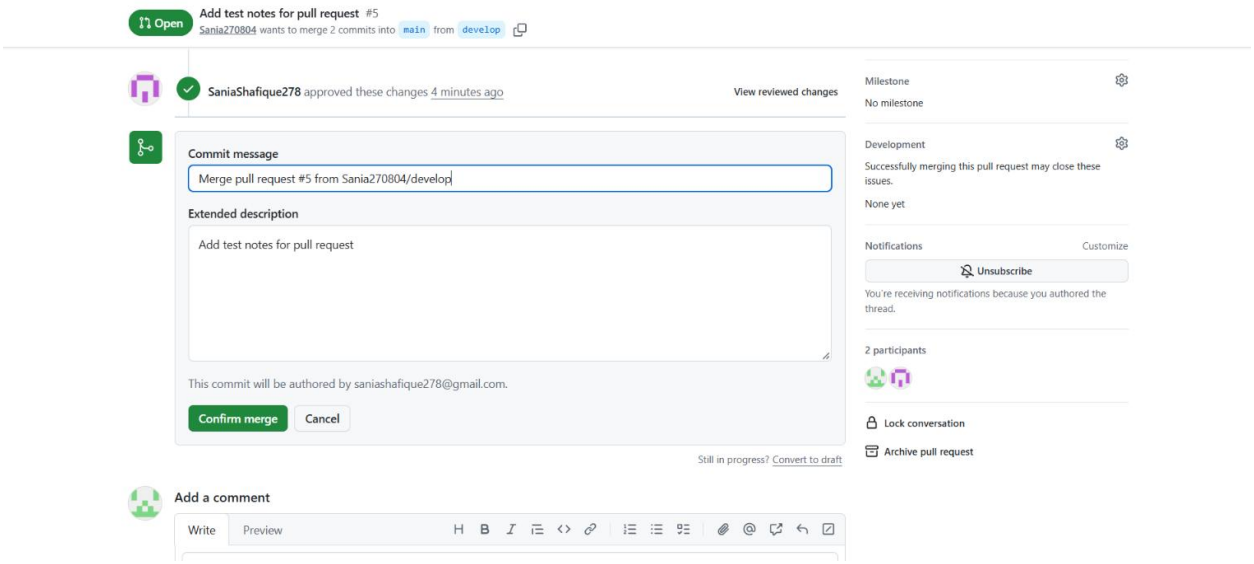
Request changes and add comments.



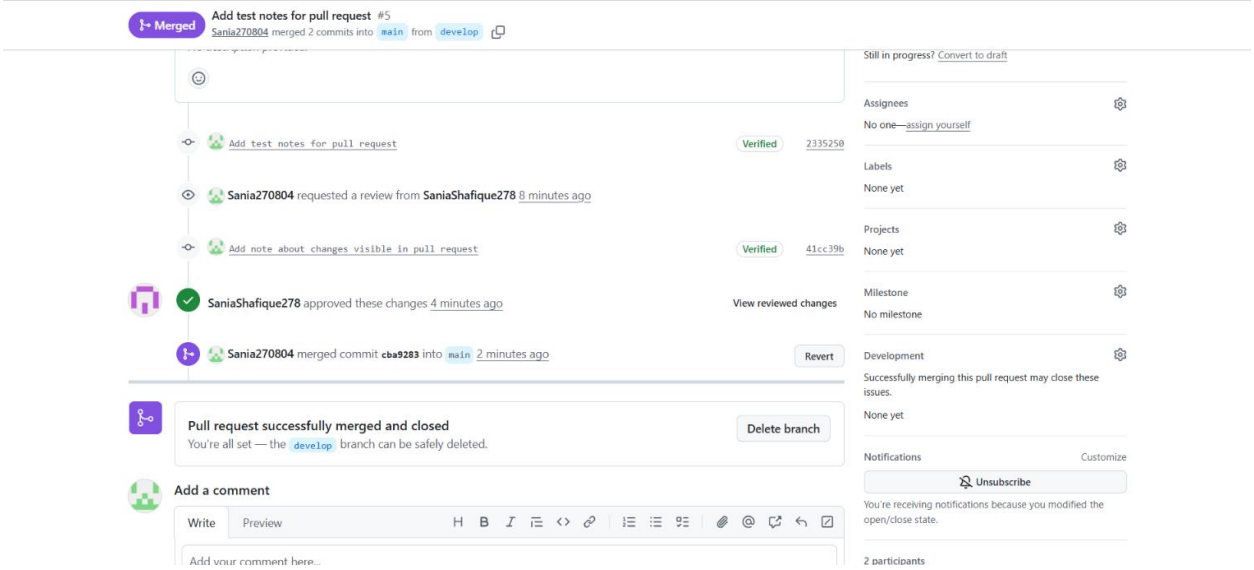
Reject or close a practice PR.



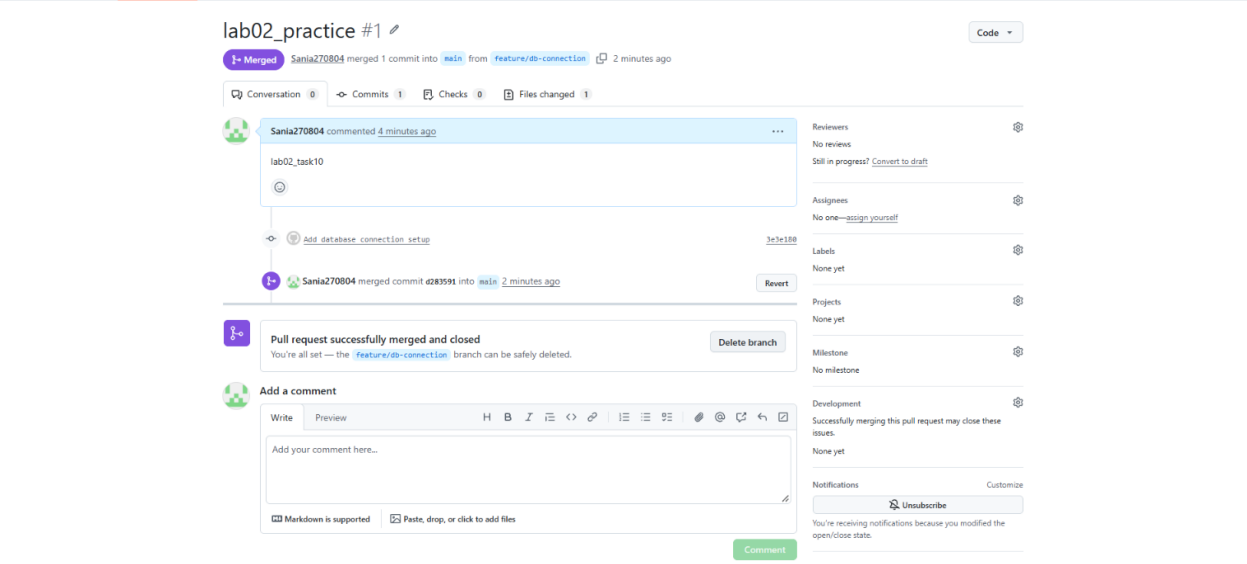
Fix issues in the same branch and push new commits.



Reviewer approves → merge using GitHub/GitLab UI.

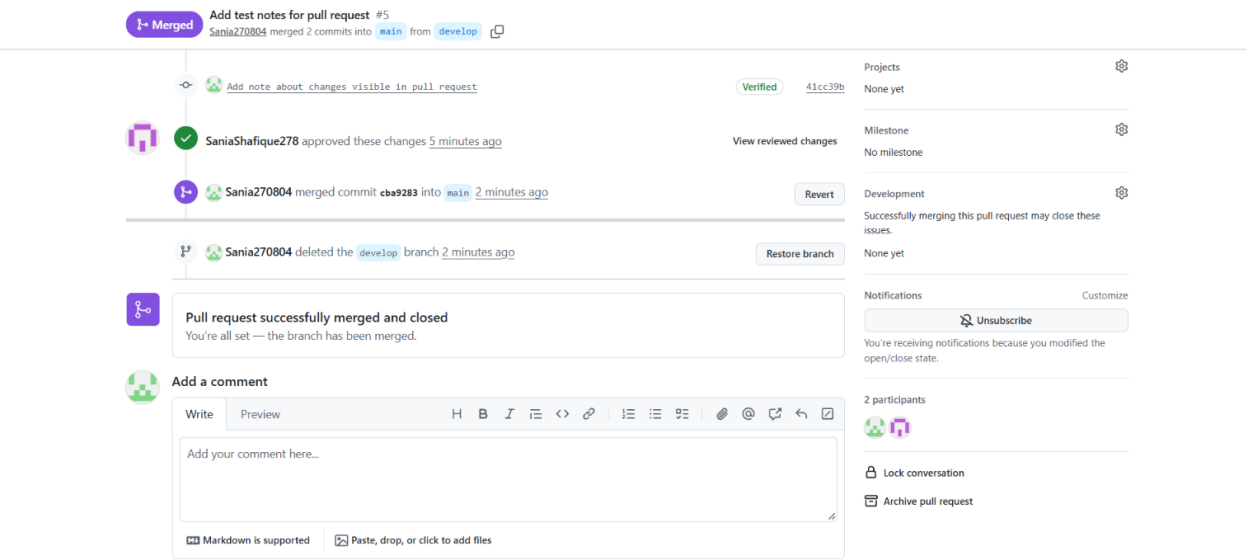


Choose merge style.



Delete the merged PR branch.

Task 13: Branch Cleanup Best Practices



Delete a merged branch through the GitHub UI.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git push origin --delete staging
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab2_Practice.git
To github.com:sania270804/DevOps_Lab2_Practice.git
 - [deleted]         staging
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Delete another completed remote branch from the terminal.

Bonus Task: Simulated Team Collaboration

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ cat notes.txt
Adding some test notes
making pull request
changes visible in pull request
Sania Shafique: I can memorize a complex keyboard shortcut instantly, but I forget why I walked into a room.
Sania Shafique: I can memorize a complex keyboard shortcut instantly, but I forget why I walked into a room.
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch collab
nothing to commit, working tree clean
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git log -n 3
commit 2cc9ebf5e4db5c1a241c774923e91d0a6f36fc36 (HEAD -> collab, origin/collab)
Author: Sania Shafique <saniashafique@gmail.com>
Date:   Sun Oct 11 12:26:28 2026 +0000

    Add Contributor name and fun fact

commit b36c76616cac1886aab71f6eb9334211c117b6a1 (main)
Author: Sania Shafique <saniashafique@gmail.com>
Date:   Sun Oct 11 12:24:08 2026 +0000

    Add Contributor name and fun fact

commit cba92833c4ed286f4377f361838131dc6a771e49 (origin/main)
Merge: 5fe7ae1 41cc39b
Author: Sania Shafique <saniashafique278@gmail.com>
Date:   Sun Oct 11 17:07:45 2026 +0500

    Merge pull request #5 from Sania270804/develop

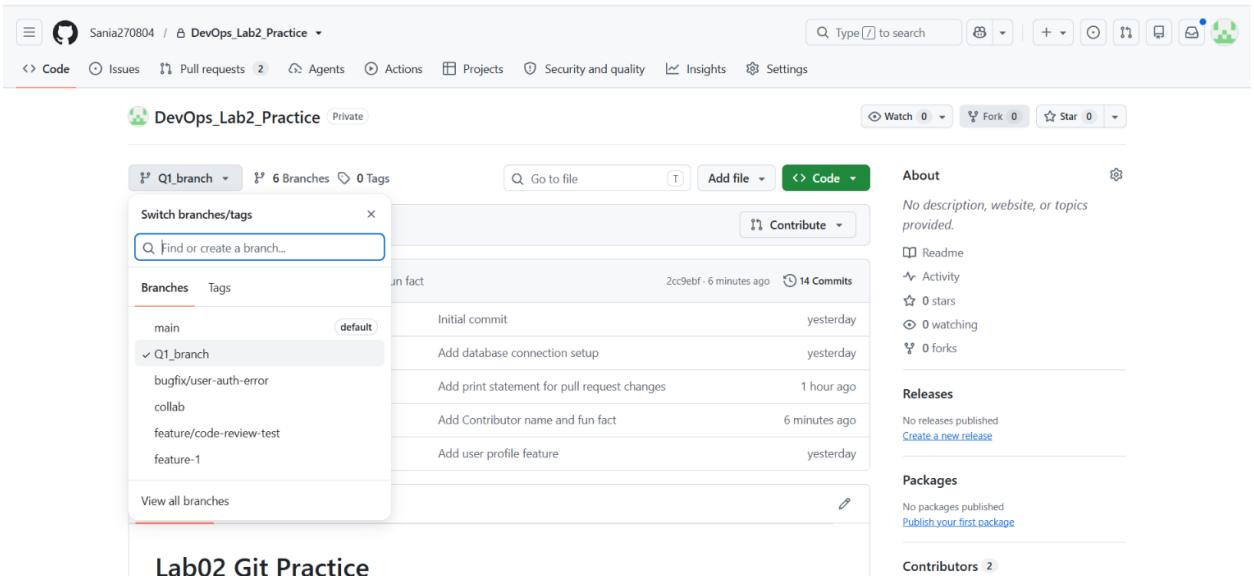
    Add test notes for pull request
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Create a branch collab. Add a collaborator on your GitHub repository. Both users should pull the latest changes, add their name and a fun fact to notes.txt, commit, and push to collab.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
git merge collab
git push origin main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
Updating b36c766..2cc9ebf
Fast-forward
 notes.txt | 1 +
 1 file changed, 1 insertion(+)
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab2_Practice.git
To github.com:sania270804/DevOps_Lab2_Practice.git
   cba9283..2cc9ebf  main -> main
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Merge collab into main and push.
Exam Evaluation Questions

1. **Advanced Branching & Merge Verification**



Create a new branch in your repository.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout Q1_branch
branch 'Q1_branch' set up to track 'origin/Q1_branch'.
Switched to a new branch 'Q1_branch'
sania270804@ubuntu:~/DevOps_Lab2_Practice$ echo "Changes made in Q1_branch" >> notes.txt
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch Q1_branch
Your branch is up to date with 'origin/Q1_branch'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   notes.txt

no changes added to commit (use "git add" and/or "git commit -a")
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Update notes.txt in Q1_branch"
[Q1_branch ef2d343] Update notes.txt in Q1_branch
 1 file changed, 1 insertion(+)
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Make a small change in a file and commit it.

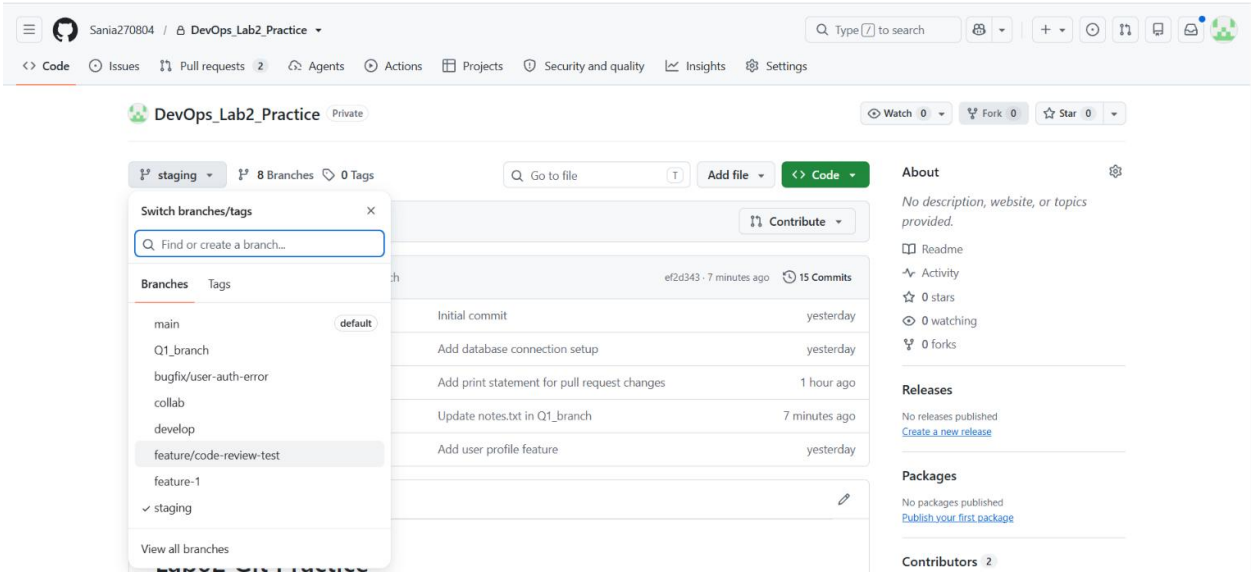
```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git pull origin main
From github.com:sania270804/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Already up to date.
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git merge Q1_branch
Updating 2cc9ebf..ef2d343
Fast-forward
 notes.txt | 1 +
 1 file changed, 1 insertion(+)
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 330 bytes | 330.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab2_Practice.git
To github.com:sania270804/DevOps_Lab2_Practice.git
   2cc9ebf..ef2d343  main -> main
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Merge this branch back into the main branch.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git log -n 5 --oneline
ef2d343 (HEAD -> main, origin/main, origin/HEAD, Q1_branch) Update notes.txt in Q1_branch
2cc9ebf (origin/collab, origin/Q1_branch, collab) Add Contributor name and fun fact
b36c766 Add Contributor name and fun fact
cba9283 Merge pull request #5 from Sania270804/develop
41cc39b Add note about changes visible in pull request
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Show the history of commits in a way that verifies the merge.

2. **Multi-Stage Workflow Simulation**



Set up a branching workflow with three branches: main, develop, and staging.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout develop
Switched to branch 'develop'
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git pull origin develop
From github.com:sania270804/DevOps_Lab2_Practice
 * branch            develop    -> FETCH_HEAD
 * [new branch]      develop    -> origin/develop
Updating 847b0dd..ef2d343
Fast-forward
 main.py   | 2 ++
 notes.txt | 5 +++++
 2 files changed, 7 insertions(+)
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature_branch
Switched to a new branch 'feature_branch'
sania270804@ubuntu:~/DevOps_Lab2_Practice$ echo "Feature branch update" >> notes.txt
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add changes on feature branch"
[feature_branch 8fddc6a] Add changes on feature branch
 1 file changed, 1 insertion(+)
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Create a feature branch from develop, make changes, and commit them.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout develop
Switched to branch 'develop'
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git merge feature_branch
Updating ef2d343..8fddc6a
Fast-forward
 notes.txt | 1 +
 1 file changed, 1 insertion(+)
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git push origin develop
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 322 bytes | 107.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab2_Practice.git
To github.com:sania270804/DevOps_Lab2_Practice.git
   ef2d343..8fddc6a  develop -> develop
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Merge the feature branch into develop.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout staging
Switched to branch 'staging'
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git pull origin staging
From github.com:sania270804/DevOps_Lab2_Practice
 * branch            staging    -> FETCH_HEAD
 * [new branch]      staging    -> origin/staging
Updating 847b0dd..ef2d343
Fast-forward
 main.py   | 2 ++
 notes.txt | 5 +++++
 2 files changed, 7 insertions(+)
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git merge develop
Updating ef2d343..8fddc6a
Fast-forward
 notes.txt | 1 +
 1 file changed, 1 insertion(+)
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin staging
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab2_Practice.git
To github.com:sania270804/DevOps_Lab2_Practice.git
   ef2d343..8fddc6a  staging -> staging
branch 'staging' set up to track 'origin/staging'.
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Merge develop into staging.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git pull origin main
From github.com:sania270804/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Already up to date.
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git merge staging
Updating ef2d343..8fddc6a
Fast-forward
 notes.txt | 1 +
 1 file changed, 1 insertion(+)
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab2_Practice.git
To github.com:sania270804/DevOps_Lab2_Practice.git
   ef2d343..8fddc6a  main -> main
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Merge staging into main.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git log --graph --oneline --all -n 10
* 8fddc6a (HEAD -> main, origin/staging, origin/main, origin/develop, origin/HEAD, staging, feature_branch, develop) Add changes on feature branch
* ef2d343 (Q1_branch) Update notes.txt in Q1_branch
* 2cc9ebf (origin/collab, origin/Q1_branch, collab) Add Contributor name and fun fact
* b36c766 Add Contributor name and fun fact
*   cba9283 Merge pull request #5 from Sania270804/develop
|\
| * 41cc39b Add note about changes visible in pull request
| * 2335250 Add test notes for pull request
* |   5fe7ae1 Merge pull request #2 from Sania270804/feature/code-review-test
|\ \
| |/
|/|
| * e9a570d (origin/feature/code-review-test) Add print statement for pull request changes
| | * 2aff9d2 (origin/feature-1) Fix typo in notes.txt
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Each stage contains the updated changes before it reaches main.

3. **Collaboration & Conflict Resolution**

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
git checkout -b collab-branch1
echo "Contributor 1: Added user authentication logic." > notes.txt
git add notes.txt
git commit -m "Contributor 1 changes to notes.txt"
Already on 'main'
Your branch is up to date with 'origin/main'.
Switched to a new branch 'collab-branch1'
[collab-branch1 9e0b15d] Contributor 1 changes to notes.txt
 1 file changed, 1 insertion(+), 7 deletions(-)
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
git checkout -b collab-branch2
echo "Contributor 2: Added database connection setup." > notes.txt
git add notes.txt
git commit -m "Contributor 2 changes to notes.txt"
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Switched to a new branch 'collab-branch2'
[collab-branch2 edc1658] Contributor 2 changes to notes.txt
 1 file changed, 1 insertion(+), 7 deletions(-)
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git branch -v
  Q1_branch               ef2d343 [ahead 1] Update notes.txt in Q1_branch
  collab                  2cc9ebf Add Contributor name and fun fact
  collab-branch1          9e0b15d Contributor 1 changes to notes.txt
* collab-branch2          edc1658 Contributor 2 changes to notes.txt
  develop                 8fddc6a Add changes on feature branch
  feature/code-review-test 625e9b9 [ahead 1, behind 1] Fix issues requested in code review
  feature/db-connection   3e3e180 Add database connection setup
  feature/user-profile    847b0dd Add user profile feature
  feature_branch          8fddc6a Add changes on feature branch
  main                    8fddc6a Add changes on feature branch
  staging                 8fddc6a Add changes on feature branch
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Work with a collaborator: both contributors should modify the same file but in separate branches.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
git merge collab-branch1
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Updating 8fddc6a..9e0b15d
Fast-forward
 notes.txt | 8 +-------
 1 file changed, 1 insertion(+), 7 deletions(-)
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git merge collab-branch2
Auto-merging notes.txt
CONFLICT (content): Merge conflict in notes.txt
Automatic merge failed; fix conflicts and then commit the result.
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Attempt to merge the branches and capture the conflict.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ nano notes.txt
sania270804@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
git commit -m "Resolve merge conflict by preserving contributions from both branches"
[main 4ee0261] Resolve merge conflict by preserving contributions from both branches
sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Resolve the conflict so that both contributions are preserved.

```
sania270804@ubuntu:~/DevOps_Lab2_Practice$ cat notes.txt
Contributor 1: Added user authentication logic.
Contributor 2: Added database connection setup.

sania270804@ubuntu:~/DevOps_Lab2_Practice$
```

Provide evidence that the final version of the file contains both collaborators' changes.